Dr Caitlin D. Kuempel

Senior lecturer · Griffith University, Brisbane, Australia
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Professional Highlights

1. Proven conservation program leader with over a decade of experience delivering applied research and management partnerships across the Pacific and Australia, focused on marine spatial planning, coral reef conservation, watershed management, and sustainable development.
2. Track record in project design and delivery: secured and managed over $1.2 million in project funding in the past two years, leading multi-institutional collaborations with governments, NGOs, and industry.
3. Strong implementation and coordination experience: served as Monitoring and Data Management Lead for WWF's Coral Reef Rescue Initiative and currently lead the Cumulative Impacts Workstream for the Blue Economy Cooperative Research Centre Marine Spatial Planning project.
4. Skilled communicator and collaborator: experienced in engaging diverse stakeholders—from local communities to global NGOs—to co-develop data-driven conservation solutions.
5. Capacity building and leadership: trained over 300 professionals and students in R programming and decision-support tools such as Marxan; mentor to early-career scientists and conservation practitioners.
6. Recognised communicator and advocate: research has been featured in The Guardian, The Washington Post, Time magazine, and The Conversation (>200,000 reads); finalist in national science communication competitions and Queensland Women in STEM Award 2025.
7. Committed to inclusive leadership and organisational excellence: lead of the Sustainable Seas Research group, active participant in Griffith's Women in Leadership Program and representative on the School's Early Career Researcher Committee.

Core competencies

Program design, delivery, and management · Partnership development · Donor and grant management · Monitoring and evaluation · Marine spatial planning · Climate and biodiversity integration · Ridge-to-reef management · Written and Oral communication to diverse audiences · Stakeholder engagement and capacity building · Strategic planning

Education

PhD Conservation Science, The University of Queensland, St. Lucia, QLD, Australia — Sept. 2018

Thesis: Evaluating trade-offs between quantitative and qualitative protected area objectives
Supervisors: Prof. Hugh Possingham, Prof. Catherine Lovelock and Dr. Alienor Chauvenet
Dean's Award for Outstanding HDR thesis (ranked within top 10% in field by examiners).

M.S. Marine Biology Northeastern University, Nahant, MA — Dec. 2013

Coral Reef Ecology (GPA 4.0/4.0), Supervisor: Dr Andrew ALtieri

BSc Environmental Science (Honours), BA French, University of Iowa, Iowa City, IA — May 2012

Honors in coral morphometrics, Certificate in Sustainability

Relevant professional Experience

Senior Lecturer, Marine Ecosystem Modelling, Griffith University, Brisbane, Australia June 2022-present

Lead the Sustainable Seas Research group, managing a portfolio of applied projects (<AUD$1.2 million) with NGOs, government and industry

Monitoring and Data Management Lead, WWF Coral Reef Rescue Initiative Feb. 2020-May 2022

Led design and rollout of monitoring systems across 7 countries and 8 partners to inform climate-resilient reef management Contributed to successful Global Environmental Facility (US$7M) proposal to support project implementation

Post-doctoral Fellow, NCEAS, Santa Barbara, CA Sept. 2018-Dec. 2019

Quantified, compared and mapped the production and environmental impacts of all major food systems as part of a collaborative working group

Conservation Planning Consultant 2018-present

I have served as a consultant in conservation planning for a variety of organisations including the IUCN, World Wildlife Fund (Fiji and Indonesia), The Nature Conservancy (Hawaii), Wildlife Conservation Society (Fiji) UNEP and Australia Awards (Papua New Guinea), among others.

Select Conservation Projects

6. Australia Awards Nature-based solutions and climate change short-course (2025): Developed a 2 week short course for 30 Papua New Guinean nations on nature-based solutiosn and climate change, delivered in country.
5. Fiji Great Sea Reef Modelling (2025): Developed sediment run-off models to guide reef protection under climate stress as part of a Green Climate Fund proposal
4. SPREP Kiwa Initiative (2024): Led micro-credential development for Pacific nature-based solutions training program.
3. Blue Economy Cooperative Research Centre Marine Spatial Planning Framework (2023–2025): Co-led national guidance on cumulative impact assessment and cross-sector coordination for sustainable ocean use in Australia.
2. Watersheds Interventiosn for Systems Health (2022-2023): Led ridge-to-reef modelling of nitrogen pollution from sanitation to inform Wildlife Conservation Society program, which was delivered to the Fijian government to inform decision making
1. WWF Coral Reef Rescue Initiative (2020–2022): Designed monitoring systems for reef health and community resilience across 7 countries and 40 partners; outputs informed Global Coral Reef Rescue strategy including a successful Global Environment Facility bid.

Media and Science Communication

I have over 10 years experience in science communiation for diverse audiences.

I have 46 peer-reviewed publications (h-index 22), 17 as lead author, >1700 citations. International collaboration on ~85% of publications. See Google scholar or ORCID

As an example of reaching a broad non-scientific audience, my written work for The Conversation blog (theconversation.com) has had >200,000 readers.

Queensland prawn areas at risk of inundation by rising seas this century The Conversation

Which food is better for the planet? The Washington Post

Food for thought: the carbon footprint of salmon and chicken farming mostly stems from feed. The Guardian

To lower food emissions consider what your dinner ate. Popular Science

Coming out or not, earthquake aid, and what's the most eco-friendly meat? Triple J Hack

Other publications, reports and white papers

4. Griffiths, Breakey, Bossi, Cossu, Dunstan, Hasan,**Kuempel, C.D.**, Lacharite, Marsh, Skinsley, Smart, Smith, Stockbridge, Woolley, Frid (2025). Implementing an Australian Marine Spatial Planning Framework. *Blue Economy Cooperative Research Centre*
3. Stockbridge and **Kuempel, C.D.** (2024). Evaluation of potential for interaction and/or conflict among ocean users and the environment in offshore waters in Australia *Blue Economy Cooperative Research Centre*
2. **Kuempel, C.D.** (2023) Sedimentation sifted out of global pollution priorities. *Science* (IF = 54.5)
1. Wenger, Jaurez, Thomas, Amaya, Corbin, Edmond, Falinski, Hill, Jenkins, Jupiter, **Kuempel, C.D.**, Lamb, Nalley, Omwenga, Oza (2023). A guide for integrated conservation and sanitation programs and approaches.*Wildlife Conservation Society*

Awards

Queensland Women in STEM Award Finalist (1 of 14 finalists) (2025)

Foundation for the WA Museum Fame Lab Science Communicaiton Semi-finalist: To lower your environmental footprint consider what your dinner ate (2023)

University of Queensland Faculty of Science Awards Rising Star - Academic category Finalist (Dec. 2020)

APEC Science Prize for Innovation, Research and Education Award - shortlisted for Australian nominee (2020)

Leadership and Outreach Roles

New Zealand Endeavor Fund Offshore wind working group - international advisory group

International Congress on Conservation Biology 2025 *Steering Committee Member*

School of Environment and Science *Early Career Representative* (2024-present)

Griffith Women in Leadership 2023 Cohort

Centre for Biodiversity and Conservation Science Management Committee *Early Career Researcher Education Lead* (2021-Present)

Soapbox Science Brisbane co-organiser (2024-present) - community engagement to promote women in STEM during National Science Week

Certified Software Carpentries Instructor (2020-present) - specialising in introductory to advanced R programming skills and version control (e.g., GitHub).

RLadies Brisbane Oraganiser - promoting women in programming (2020-2024)

Marxan Spatial Planning Instructor (2017-current) - successful workshops in Australia, Indonesia, Colombia, and online